UNIVERSITAS PADJADJARAN - Digital Business Study Program

Statistical Analysis II - Midterm Examination (UTS) 2026/2027

Student: _____ (Name) | NPM: 120510253003 | Submitted: 13 Oct 2026 | Lecturer: Dr. Vita Sarasi, S.E., M.T.

Assigned country: Viet Nam (last digit of NPM d = 3, 4th country). **Comparison countries:** Philippines (2nd), India (3rd). Role: business analyst advising a multinational considering investment in Viet Nam. Significance level $\alpha = 0.05$ throughout.

Files: this report; workbook 120510253003_NAMA_UTS.xlsx (raw downloads + transformations + live Excel formulas for Q1-Q3); reproducible code *fetch_wb.py, analysis.py, build_workbook.py, build_report.py, q7_reliability.py*; raw data in *data/raw/*. All numbers below are produced by *analysis.py* (results.json).

Summary of findings

Q	Method	Key result ($\alpha = 0.05$)	Decision / implication
1	Lilliefors, VN CPI inflation 1996-2025 (n=30)	D = 0.211 > 0.161; p = 0.002 -> not normal	Use quantiles/bootstrap; plan for inflation tail risk
2	χ^2 2×2, unemployed vs employed, VN vs PH, 2024	rates 1.53% vs 2.36%; V = 0.030; weighted counts -> p unreliable	Tighter labour market in VN: hiring premium
3	χ^2 2×3 sector employment, VN vs PH	V = 0.22; industry 33% vs 18%	Manufacturing sourcing in VN; services staffing in PH
4	ANOVA / Kruskal-Wallis, GDP growth, 3 countries × 25 yrs	ANOVA p = 0.162; Kruskal-Wallis p = 0.017; Friedman p = 0.006; PH vs IN differ	VN growth ≈ India's with less than half the volatility
5	OLS GDP growth ~ investment + inflation (n=30) + quadratic	F p = 0.960; R^2 = 0.003; no quadratic effect	Macro predictors give no timing signal; stage investment
6	Trend vs MA(3), holdout 2021-25, exports	MA(3) MAPE 30.3% < linear 44.1%; 10-yr trend best (10.9%)	Plan on US$ 500-560 bn exports 2026-27; stage capacity
7	Reliability on public data (Zenodo, n = 248, 6 items)	$\alpha = 0.94$; Spearman-Brown = 0.95; item-total r ≥ 0.79	Reliable but possibly redundant; validity not shown
8	Executive memo	see Q8	REDESIGN: phased expansion with triggers

Data sources, all extracted 12 Oct 2026

Dataset	Exact URL / API call	Indicator code	Unit	Period
World Bank WDI - CPI inflation	https://api.worldbank.org/v2/country/VNM;PHL;IND/indicator/FP.CPI.TOTL.ZG?format=json&per;_page=20000&date;=1960:2025	FP.CPI.TOTL.ZG	% y/y	1996-2025
World Bank WDI - GDP growth	same, indicator NY.GDP.MKTP.KD.ZG	NY.GDP.MKTP.KD.ZG	% y/y	2000-2024 / 1996-2025
World Bank WDI - Gross capital formation	same, indicator NE.GDI.TOTL.ZS	NE.GDI.TOTL.ZS	% of GDP	1996-2025
World Bank WDI - Exports G&S	same, indicator NE.EXP.GNFS.CD	NE.EXP.GNFS.CD	current US$	1986-2025
World Bank WDI - context (Q8)	same, indicators NE.EXP.GNFS.ZS, EG.USE.ELEC.KH.PC	NE.EXP.GNFS.ZS; EG.USE.ELEC.KH.PC	% GDP; kWh/cap	to 2025 / 2024
ILOSTAT SDMX - unemployed	https://sdmx.ilo.org/rest/data/ILO,DF_UNE_TUNE_SEX_AGE_NB/VNM+PHL+IND....?format=csv&startPeriod;=2015	UNE_TUNE_NB, age 15+	thousand persons	2024
Zenodo - survey B (Q7)	https://zenodo.org/api/records/22057938/files/DATAUPDATE.csv/content	SCR1-SCR6	Likert 1-5	2025-26 survey
Zenodo - survey A (Q7)	https://zenodo.org/api/records/20439760/files/Firm%20Survival%20Dataset.csv/content	OR1-OR5	Likert 1-7	2025 survey
ILOSTAT SDMX - employed by sector	https://sdmx.ilo.org/rest/data/ILO,DF_EMP_TEMP_SEX_ECO_NB/VNM+PHL+IND....?format=csv&startPeriod;=2015	EMP_TEMP_NB by ECO_SECTOR_*	thousand persons	2024

Data-integrity note. Nothing was typed in or invented: World Bank JSON and ILOSTAT CSV files were downloaded programmatically and are stored unchanged in data/raw/. Viet Nam's CPI series starts in 1996, and the World Bank 2025 values are the latest available (database update 2026-10-08) and may be revised. ILOSTAT values are **weighted labour-force-survey estimates** (not raw respondent counts), which matters for Q2-Q3.

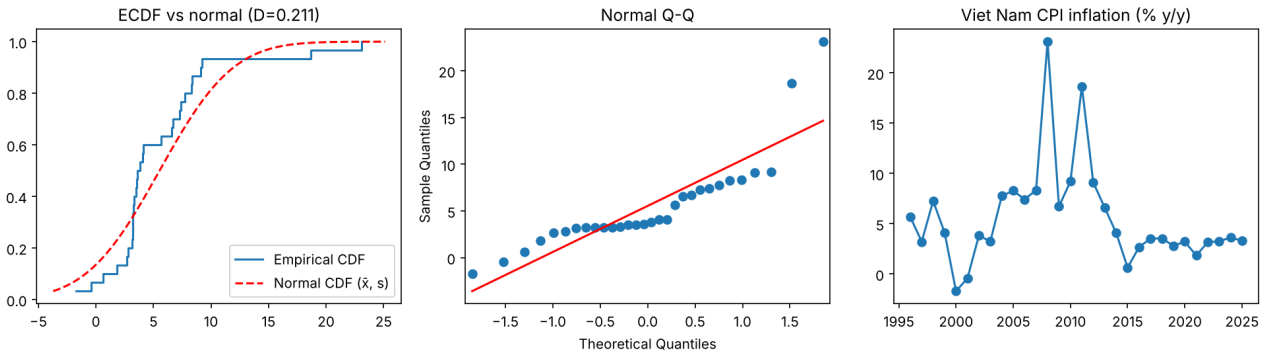
Question 1 - Normality testing (Lilliefors)

Data. Viet Nam CPI inflation (FP.CPI.TOTL.ZG), annual %, 1996-2025, n = 30 (≥ 20 required). Mean = 5.53, s = 5.02, median 3.73, min -1.71, max 23.12 (2008), skewness 2.04, excess kurtosis 5.44.

Hypotheses. H0: inflation is drawn from a normal distribution (mean and variance unknown, estimated from the sample); H1: it is not. **Test.** Lilliefors = Kolmogorov-Smirnov with parameters estimated: $D = \max_i \max\{ i/n - F(z_i), F(z_i) - (i-1)/n \}$, $z_i = (x_{(i)} - \text{mean})/s$, F = standard-normal CDF. Ordered observations, z-values, theoretical and empirical CDFs are in workbook sheet *Q1* (live formulas) and tables/Q1_lilliefors.csv.

i	x(i) %	Year	z	F theor.	i/n	D+	D-		i	x(i) %	Year	z	F theor.	i/n	D+	D-
1	-1.71	2000	-1.44	0.074	0.033	-0.041	0.074		16	3.83	2002	-0.34	0.367	0.533	0.166	-0.133
2	-0.43	2001	-1.19	0.117	0.067	-0.050	0.084		17	4.08	2014	-0.29	0.386	0.567	0.180	-0.147
3	0.63	2015	-0.98	0.164	0.100	-0.064	0.097		18	4.12	1999	-0.28	0.389	0.600	0.211	-0.178
4	1.83	2021	-0.74	0.230	0.133	-0.097	0.130		19	5.67	1996	0.03	0.511	0.633	0.122	-0.089
5	2.67	2016	-0.57	0.284	0.167	-0.117	0.151		20	6.59	2013	0.21	0.584	0.667	0.083	-0.050
6	2.80	2019	-0.55	0.293	0.200	-0.093	0.126		21	6.72	2009	0.24	0.593	0.700	0.107	-0.073
7	3.16	2022	-0.47	0.318	0.233	-0.084	0.118		22	7.27	1998	0.35	0.635	0.733	0.098	-0.065
8	3.21	1997	-0.46	0.321	0.267	-0.055	0.088		23	7.42	2006	0.38	0.646	0.767	0.120	-0.087
9	3.22	2020	-0.46	0.322	0.300	-0.022	0.056		24	7.75	2004	0.44	0.671	0.800	0.129	-0.096
10	3.23	2003	-0.46	0.323	0.333	0.010	0.023		25	8.28	2005	0.55	0.708	0.833	0.125	-0.092
11	3.25	2023	-0.45	0.325	0.367	0.042	-0.009		26	8.34	2007	0.56	0.712	0.867	0.154	-0.121
12	3.31	2025	-0.44	0.329	0.400	0.071	-0.038		27	9.09	2012	0.71	0.761	0.900	0.139	-0.106
13	3.52	2017	-0.40	0.344	0.433	0.089	-0.056		28	9.21	2010	0.73	0.768	0.933	0.165	-0.132
14	3.54	2018	-0.40	0.345	0.467	0.121	-0.088		29	18.68	2011	2.62	0.996	0.967	-0.029	0.062
15	3.62	2024	-0.38	0.351	0.500	0.149	-0.115		30	23.12	2008	3.51	1.000	1.000	0.000	0.033

Statistic	Value
D (max |F_n - F|)	0.2113
Critical value, α = 0.05, n = 30 (Lilliefors 1967 table)	0.161
Approx. p-value (statsmodels, Lilliefors table method)	0.002
Decision	D > critical value and p < 0.05 -> **reject H0**
Cross-checks: Shapiro-Wilk / Jarque-Bera	W = 0.795, p = < 0.001 / JB = 42.7, p = < 0.001
Robustness: Lilliefors on log(1+π)	D = 0.208, p = 0.002 (still rejected)
Lag-1 autocorrelation	0.44



Interpretation. Normality is **not reasonable** for Viet Nam's inflation: the empirical CDF departs from the normal CDF by D = 0.211, well above 0.161. The cause is visible: a heavy right tail (skewness 2.0) created by the 2008 spike (23.1%) and 2011 (18.7%), versus a calm 2015-2025 (0.6-3.6%). Log-transforming does not fix it. **Time-series caveat:** the test

assumes independent observations; inflation is positively autocorrelated (lag-1 r = 0.44), so the effective sample size is smaller than 30 and the stated p-value is only indicative (the table critical value is for iid data). Also, the 1996-2025 span mixes policy regimes, so 'one distribution' is itself questionable.

Implications. (i) Do not use mean ± 1.96·s to describe 'normal' inflation ranges or to build interval forecasts for inflation; use the median/quantiles or a bootstrap. (ii) Tests on means (t, ANOVA) applied directly to inflation levels are unreliable in small samples; prefer rank-based tests or transformations and model residuals, not the raw series. (iii) For budgeting, treat inflation as skewed with tail risk (price-escalation clauses) rather than symmetric noise.

Question 2 - Chi-square test of two proportions

Data and outcome. ILOSTAT, 2024, persons aged 15+: unemployed (UNE_TUNE_NB) and employed (EMP_TEMP_NB, ECO_SECTOR_TOTAL), Viet Nam vs Philippines (counts = ILO thousands × 1000). Binary outcome: unemployed vs employed within the labour force. **H0:** the unemployment proportion is equal in the two countries ($p_{VN} = p_{PH}$); H1: they differ.

Observed	**Unemployed**	**Employed**	**Total**	Unemployment rate
Viet Nam	858,467	55,280,158	56,138,625	1.53%
Philippines	1,147,686	47,524,318	48,672,004	2.36%

Expected = row total × col total / N	**Unemployed**	**Employed**
Viet Nam	1,074,534.8	55,064,090.2
Philippines	931,618.2	47,740,385.8

Result. $\chi^2 = \Sigma(O-E)^2/E = 95{,}384.8$, df = (2-1)(2-1) = 1, $p < 0.001$ (Yates-corrected $\chi^2 = 95{,}384.3$); all expected counts > 5. Effect size: Cramér's V = 0.030 (very small); difference in rates = -0.83 pp.

Why this p-value must not be taken at face value. These counts are *weighted survey estimates* grossed up to national populations (N ≈ 105 million), not the number of people actually interviewed. A chi-square test treats each count as an independent observation, so with ~105 million 'observations' it will flag even trivial differences, and it ignores the survey's clustering, stratification and weighting (design effect). Unweighted respondent counts were not available from ILOSTAT; the correct approach is a survey-adjusted (Rao-Scott) test on microdata. As a transparent sensitivity analysis I re-ran the test keeping the observed proportions but assuming **illustrative** effective sample sizes per country (these n are assumptions, not data):

Effective n per country (assumed)	χ^2	**p-value**
2,000	3.60	0.058
10,000	18.02	< 0.001
50,000	90.11	< 0.001
200,000	360.43	< 0.001

Interpretation. The *direction* (Viet Nam lower, about 0.8 pp) is stable, but whether it is statistically significant depends on the unknown effective sample size (non-significant at n_{eff} = 2,000, significant from ~10,000). I therefore conclude that Viet Nam's measured unemployment rate is lower than the Philippines' in 2024, with a small practical effect, and that formal significance cannot be established from these aggregates.

Management implication. Both labour markets are tight by ILO's definition, but Viet Nam's lower unemployment suggests a smaller pool of immediately available workers and more wage/retention pressure than the Philippines. If expansion needs rapid hiring of non-specialised staff, budget a recruitment premium and training pipeline in Viet Nam; the Philippines offers more slack. **Limitations:** unemployment omits informality and underemployment; one year; different survey designs.

Question 3 - Chi-square test of independence (2 × 3)

Data. ILOSTAT EMP_TEMP_NB by broad sector (ECO_SECTOR_AGR / IND / SER), 2024, both sexes, Viet Nam vs Philippines. **H0:** the sectoral distribution of employment is independent of country; H1: it is not. Viet Nam has 3,845,332 workers (6.96%) 'not classified by sector' (ECO_SECTOR_X), excluded from the main test; a 2×4 version including them is given as robustness.

Observed (persons)	**Agriculture**	**Industry**	**Services**	**Total**
Viet Nam	13,637,057	17,131,226	20,666,543	51,434,826
Philippines	9,858,510	8,649,772	29,016,036	47,524,318

Expected under H0	**Agriculture**	**Industry**	**Services**
Viet Nam	12,212,013	13,399,885	25,822,928
Philippines	11,283,554	12,381,113	23,859,651
Row % Viet Nam / Philippines	26.5% / 20.7%	33.3% / 18.2%	40.2% / 61.1%
Std. residuals VN	408	1,019	-1,015

Result. χ^2 = 4,653,827, df = (2-1)(3-1) = 2, p < 0.001; Cramér's V = 0.217 (moderate). Robustness 2×4 incl. 'not classified': χ^2 = 8,107,444, df = 3, V = 0.281 - same conclusion. **Small cells:** minimum expected count = 11,283,554 persons - no small-cell problem (rule: all E ≥ 5). The goodness-of-fit alternative (Viet Nam counts vs Philippine shares as 'benchmark') is also rejected (χ^2 = 10,944,882), but I treat that only as a descriptive check because the Philippines is not a justified published benchmark.

Interpretation. Sector mix clearly depends on country: Viet Nam is far more industry-oriented (33.3% vs 18.2% of classified jobs) and more agricultural (26.5% vs 20.7%), whereas the Philippines is services-dominated (61.1% vs 40.2%). As in Q2, the p-value is mechanically ~0 because the counts are weighted national estimates; the **effect size** (V ≈ 0.22, standardised residuals above ±400) is the informative quantity, and the pattern is consistent with well-known structural differences.

Management decision (sourcing / staffing). Locate labour-intensive manufacturing and light-assembly sourcing in Viet Nam, where industry employs 17.1 million workers and a manufacturing labour pool exists; place business-process and customer-service staffing in the Philippines (services share 61%). Verify local skills at city level before committing, since national shares do not describe a specific industrial zone. **Limitation:** one year, broad sectors, Viet Nam's 7% unclassified.

Question 4 - One-way ANOVA across three countries

Panel. Annual GDP growth (NY.GDP.MKTP.KD.ZG, %), Viet Nam, Philippines, India, matching years 2000-2024 (25 per country, N = 75; sheet *Q4_panel*). **H0:** $\mu_{VN} = \mu_{PH} = \mu_{IN}$; H1: at least one differs.

Country	Mean	Median	SD	Shapiro-Wilk W (p)
VNM	6.33	6.69	1.35	0.861 (0.003)
PHL	4.98	5.71	3.36	0.595 (< 0.001)
IND	6.20	7.21	3.03	0.721 (< 0.001)

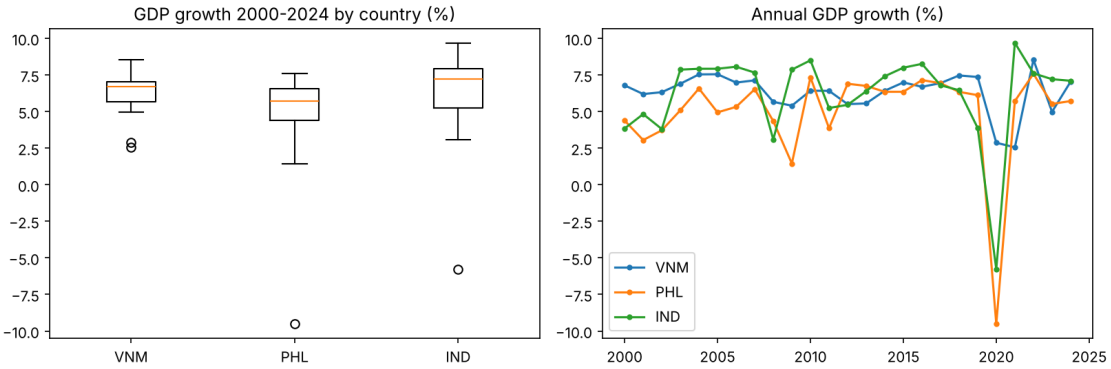
Assumption checks. Normality within groups is **rejected** for all three countries (Shapiro p < 0.01; skewed by crisis years such as 2020). Homogeneity of variance: Levene (median-centred) F = 1.12, p = 0.333 does not reject, but Bartlett χ^2 = 18.2, p = < 0.001 does, and the SDs differ visibly (1.35 vs 3.36 vs 3.03). Because Bartlett is itself sensitive to non-normality and Levene has modest power with n = 25, I treat variances as unequal and normality as violated; hence the rank-based Kruskal-Wallis test is designated the primary test, with classical and Welch ANOVA reported for completeness.

Test	Statistic	df	p-value
Classical one-way ANOVA	F = 1.865	2, 72	0.162
Welch ANOVA (unequal variances)	F = 1.710	2, 40.6	0.194
Kruskal-Wallis (primary)	H = 8.173	2	0.017

ANOVA table: $SS_{between}$ = 27.76, SS_{within} = 535.80, $MS_{between}$ = 13.88, MS_{within} = 7.44, η^2 = 0.049.

Post-hoc (justified because Kruskal-Wallis is significant). Dunn's test with tie correction and Holm adjustment (rank-based, consistent with the primary test); Tukey HSD on means is shown for reference since the ANOVA F was not significant.

Pair	Mean-rank diff.	z	p raw	p Holm	Tukey p (means)
VNM-PHL	11.48	1.86	0.063	0.125	0.195
VNM-IND	-5.84	-0.95	0.343	0.343	0.986
PHL-IND	-17.32	-2.81	0.005	0.015	0.259



Interpretation. The classical and Welch ANOVAs find **no significant difference in mean growth** (means 6.33, 4.98, 6.20%), but the rank-based test does (mean ranks IN 45.7 > VN 39.9 > PH 28.4). I do not pick the test I like: because normality fails, Kruskal-Wallis is the appropriate one, and it indicates the distributions differ - driven by **Philippines vs India** (Holm p = 0.015); Viet Nam is not significantly different from either after Holm adjustment. Kruskal-Wallis compares distributions (location and spread), not means, so it does not prove that mean growth differs. Sensitivity: excluding 2020-2021 (COVID) the classical ANOVA becomes significant (F = 4.15, p = 0.020; means VN 6.64, PH 5.58, IN 6.57) - so the conclusion is sensitive to the pandemic years.

Independence limitation. Treating successive years as independent is questionable. Within-country lag-1 autocorrelation of growth is small (VN 0.08, PH -0.04, IN 0.03), but the three countries are **not independent groups**: the same years (2009, 2020) hit all of them, so the design is really repeated measures by year (a Friedman/mixed-model analysis would be more appropriate), and the 25 years are one realisation of a structurally changing economy, not

random draws. Hence p-values are indicative. As the more appropriate repeated-measures check I ran a Friedman test with the 25 years as blocks: $\chi^2(2) = 10.32$, $p = 0.006$ (mean ranks within year: VN 2.20, PH 1.48, IN 2.32), which agrees with Kruskal-Wallis: the Philippines ranks lowest in a typical year.

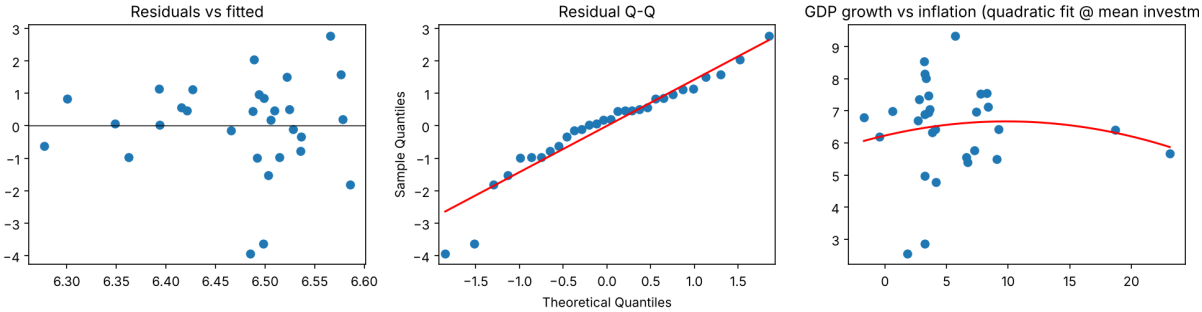
Recommendation. Viet Nam delivered average growth equal to India's with less than half the volatility (SD 1.35 vs 3.03 pp), which favours it for planning stability; do not claim it grows faster than India. The Philippines' growth distribution is weaker than India's.

Question 5 - Multiple and nonlinear regression

Data. Viet Nam, 1996-2025, n = 30 aligned annual observations (sheet *Q5_data*). Y = GDP growth (NY.GDP.MKTP.KD.ZG, %); X1 = gross capital formation (NE.GDI.TOTL.ZS, % GDP); X2 = CPI inflation (FP.CPI.TOTL.ZG, %). Predictors were chosen a priori from growth theory (investment drives capacity; inflation proxies macro-stability/energy-cost pressure), not by searching for significance. Model: $Y = \beta0 + \beta1 \cdot X1 + \beta2 \cdot X2 + \varepsilon$. H0 (each coefficient): $\beta = 0$; model F-test H0: $\beta1 = \beta2 = 0$.

Term	Coef.	Std. err.	t	p-value	HAC p
Intercept	7.212	3.334	2.16	0.040	0.032
Investment (X1)	-0.022	0.106	-0.20	0.840	0.830
Inflation (X2)	-0.006	0.061	-0.10	0.921	0.879

Fit. F(2, 27) = 0.041, p = 0.960; R^2 = 0.0030, adjusted R^2 = -0.071; residual SE = 1.50 pp. **Diagnostics.** Multicollinearity: VIF = 1.19 for both predictors (corr(X1,X2) = 0.40) - not a problem. Heteroskedasticity: Breusch-Pagan LM = 1.33, p = 0.514 (no evidence). Autocorrelation: Durbin-Watson = 1.53, Breusch-Godfrey(1) p = 0.376 (no evidence; HAC standard errors give the same inference). Normality of residuals: Shapiro-Wilk W = 0.935, p = 0.068 (borderline pass). Largest Cook's distance = 0.18 (1996; < 1, no influential point). The diagnostics are acceptable - the problem is not a violated assumption but an **absence of signal**.



Result. Neither predictor is significant (p = 0.840 and 0.921), the F-test does not reject H0, and $R^2 \approx 0.003$: investment share and inflation explain essentially none of year-to-year GDP-growth variation in Viet Nam. Excluding 2020-2021 gives the same picture (R^2 = 0.049).

Nonlinear extension. Quadratic term added to inflation (centred, to limit collinearity) and, separately, to investment; compared by nested F-test, adjusted R^2 and AIC:

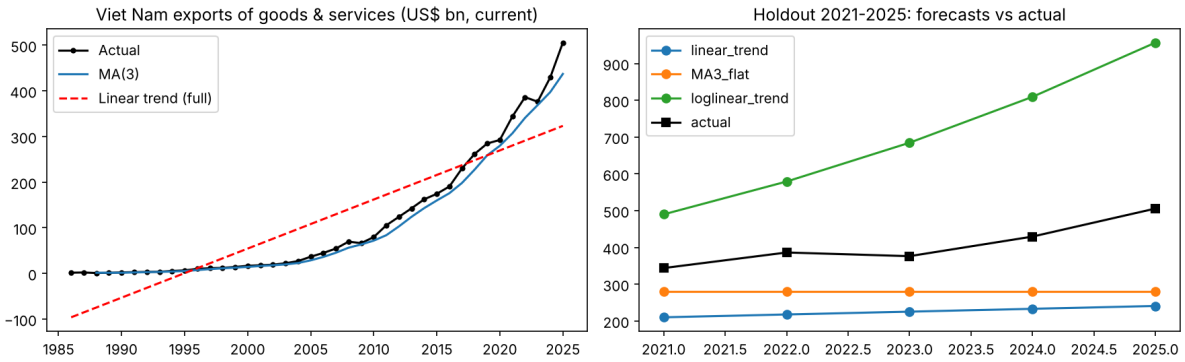
Model	R^2	Adj. R^2	AIC	Quadratic term: coef (p)	Nested F-test p
Linear (X1, X2)	0.0030	-0.071	112.5	-	-
+ Inflation²	0.0186	-0.095	114.0	-0.0045 (0.526)	0.526
+ Investment²	0.0199	-0.093	114.0	0.0186 (0.510)	0.510

Interpretation. Neither quadratic term is significant, adjusted R^2 falls and AIC rises: the linear model is preferred, and even it explains nothing. **Association is not causation:** even a significant coefficient here would reflect omitted variables (exports/FDI, global demand, policy, the COVID shock), reverse causality (high growth attracts investment) and the shared time trend, with n = 30 and low power. A null result also does not prove that investment or inflation do not matter, only that these two annual aggregates do not predict growth in this sample.

Recommendation. Do not time investment or set prices using these macro predictors. Stage the capital commitment against firm-level demand evidence, and protect margins with price-adjustment (indexation) clauses because inflation volatility (Q1) is a cost risk even if it does not predict aggregate growth in the data.

Question 6 - Time-series forecasting

Series. Viet Nam exports of goods and services (NE.EXP.GNFS.CD), current US$ billion, annual, 1986-2025, n = 40 (≥ 10). It is business-relevant for supply-chain, procurement and capacity planning. **Components.** Trend: strong, rising from a few US$ bn in the 1980s to US$ 506 bn; mean annual growth 17.3% (SD 17.9 pp), i.e. closer to exponential early and decelerating later. Cycle: contractions in 2009 (-5%) and 2023 (-2.6%), slowed growth in 2020 (+3%) and a rebound in 2021 (+18%). Irregular movements: nominal-price and exchange-rate effects. **Seasonality: not identifiable** - annual data contain no within-year information, so no seasonal claim is made.



Method. Time-ordered holdout: models fitted on 1986-2020 and evaluated on the last 5 years (2021-2025) without shuffling. Pre-specified models: (a) linear trend y = b0 + b1·t (training slope 7.70 bn/yr); (b) 3-year moving average (multi-step forecast = last 3-year mean). Extensions added after seeing that (a) and (b) lag a still-growing series: linear trend on the last 10 training years, damped Holt trend, log-linear trend and a naive last-value benchmark. A rolling one-step MA(3) is also shown but is *not comparable* with multi-step forecasts.

Model (holdout 2021-2025)	MAE (US$ bn)	**MAPE (%)**	**RMSE**
Linear trend (full training) [pre-spec.]	182.9	44.1	188.5
MA(3), 5-step flat [pre-spec.]	128.6	30.3	140.2
Linear trend, last 10 yrs [ext.]	46.9	10.9	55.3
Damped Holt [ext.]	60.4	14.1	68.9
Naive last value [ref.]	115.8	27.1	128.6
Log-linear trend [ext.]	295.7	70.4	316.7
MA(3) rolling 1-step [not comparable]	69.6	16.8	73.6

MAPE is valid because the series is strictly positive. **Result:** of the two pre-specified models, the moving average beats the linear trend (MAE 128.6 vs 182.9; MAPE 30.3% vs 44.1%) - but both under-forecast badly because they lag a rising series. The extension 'linear trend on the last 10 years' is best (MAPE 10.9%); the log-linear trend over-forecasts (MAPE 70.4%) as growth slowed. With only 5 holdout points this ranking is itself uncertain, and the window length was chosen by judgement, so I treat it as a reasonable planning model, not a proven winner.

Forecast (US$ bn)	**2026**	**2027**	95% interval
Linear trend, full series	334	345	192-476; 202-487
MA(3)	437	437	not defined (no error model)
Linear trend, last 10 years (recommended base)	502	533	447-557; 476-591
Damped Holt	518	558	-

Uncertainty and validity warnings. The full-series linear trend forecasts US$ 334 bn for 2026, *below the 2025 actual of 506 bn*, and the 2025 actual lies outside its own 95% interval: the model is mis-specified (Durbin-Watson = 0.08, i.e. extreme residual autocorrelation, so its OLS intervals are far too narrow) and must not be used for planning. Intervals from the 10-year fit assume a stable linear trend and iid errors with 8 degrees of freedom, so they are optimistic. Values are nominal US$, so exchange-rate and price movements, tariffs and the energy-price news in the case can move them.

Business response (procurement / capacity). Plan on a base case of roughly US$ 500-560 bn exports in 2026-27 (flat to +10% vs 2025): size warehouse and logistics capacity in two stages (about 80% of the base-case volume committed,

the remainder as options/leases), hold 4-6 weeks of critical-input inventory, and use review triggers - if actual 2026 exports fall below the interval lower bound (about US$ 447 bn) pause the second stage; if they exceed US$ 557 bn, accelerate it.

Question 7 - Validity and reliability of a management survey

Construct and respondents. Perceived supply-chain resilience to energy-price shocks (news context: IMF warning on energy shocks, 7 Oct 2026; IEA EV-transition outlook, which reshapes energy and component supply). **Target respondents:** operations, procurement and supply-chain managers (≥ 2 years in role) of manufacturing, logistics and trading firms in Viet Nam, including MNC subsidiaries and SMEs.

(a) Six designed items (5-point Likert: 1 = strongly disagree … 5 = strongly agree):

#	Item	Facet / content-validity rationale
1	My firm could keep production running if energy prices stayed sharply higher for the next six months.	Operational continuity under an energy-price shock (core of the IMF warning)
2	We have qualified alternative suppliers for our most critical inputs.	Supplier diversification (pressure to diversify supply chains)
3	We could pass at least part of higher energy or transport costs to customers without losing major accounts.	Pricing power / cost pass-through
4	Our logistics network can be rerouted quickly if a key supplier or shipping route is disrupted.	Logistics flexibility
5	Our inventory buffers of critical inputs would cover at least one month of supply disruption.	Buffer capacity
6	Management regularly monitors energy-price and supplier-risk indicators and acts on them.	Sensing and response capability

Content validity. Each item maps to one mechanism in the news (energy-cost shock, supplier diversification, buffers, flexibility, response capability), all are positively worded and about the same construct (no reverse-coding needed). Planned verification before fielding: 3-5 expert ratings of relevance to compute item-CVI (retain ≥ 0.78 with ≥ 6 experts) and cognitive pretest with ~5 managers. Limitation: six items cannot cover every facet, and pricing power (item 3) is conceptually distinct from buffers (item 5), which could lower alpha.

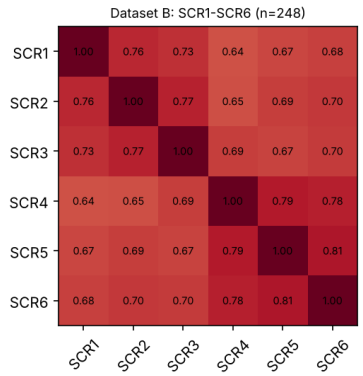
(b) Public item-level dataset used for the computations. I did not collect responses. I used two openly licensed (CC-BY 4.0) survey datasets from Zenodo; both are real respondent data, nothing simulated.

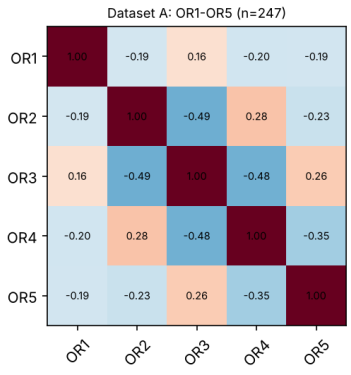
	Dataset B (primary)	Dataset A (comparison)
Record	Zenodo 10.5281/zenodo.22057938 (Yuniarty, published 22 Aug 2026), file DATAUPDATE.csv - https://zenodo.org/records/22057938	Zenodo 10.5281/zenodo.20439760 (Khan, 29 May 2026), file Firm Survival Dataset.csv - https://zenodo.org/records/20439760
Respondents / scale	n = 248; 5-point Likert	n = 247 Indonesian manufacturing/supply-chain firms (managers); 7-point scale (observed 5-7)
Items used	SCR1-SCR6 = six-item supply-chain-resilience block (k = 6, same length as my design; exactly 3 + 3 for odd-even split)	OR1-OR5 = five 'operational resilience' items with published wording (e.g. 'Our firm can quickly recover from supply chain disruptions')
Why chosen	Same construct, same item count and scale type as the exam design. **Caveat:** the deposit contains no codebook or item wording (verified: record has only the CSV), so item-by-item content equivalence with my six items cannot be confirmed.	Item wording known and construct close, but only five items and a different scale. It was examined first; dataset B was selected afterwards because it matches the design better. Both are reported.

(c) Results. Corrected item-total correlation $r_{i(T-i)}$ = Pearson r of item i with the sum of the other items; split-half: odd items (1,3,5) vs even items (2,4,6), Pearson r_{half} then Spearman-Brown $r_{SB} = 2r/(1 + r)$; Cronbach's $\alpha = k/(k-1)\cdot(1 - \Sigma s^2_i/s^2_T)$, 95% CI by 2,000 bootstrap resamples.

Item (dataset B)	**Mean**	**SD**	**% at max**	Corrected item-total r	α if deleted
SCR1	3.20	1.46	22.6	0.787	0.929
SCR2	3.33	1.44	26.6	0.812	0.926
SCR3	3.22	1.40	17.7	0.812	0.926
SCR4	3.45	1.41	27.0	0.806	0.927
SCR5	3.56	1.46	35.9	0.829	0.924
SCR6	3.46	1.32	23.0	0.840	0.923

Statistic (dataset B, n = 248, k = 6)	**Value**	**Benchmark**
Cronbach's α (95% CI)	0.938 (0.923-0.950)	≥ 0.70 acceptable; > 0.90 may signal redundancy
Mean inter-item correlation	0.716	0.15-0.50 typical; higher = redundant
Odd-even split-half r (Pearson)	0.908	-
Spearman-Brown reliability r_{SB}	0.952	≥ 0.70
Rank-based check: Spearman ρ half = / SB from ρ	0.741 / 0.851	robustness to non-linearity / ties
Sensitivity: α excl. 3 straight-liners / excl. 2 duplicate rows	0.937 / 0.937	stable





Interpretation (dataset B). All corrected item-total correlations are high (0.79-0.84, far above 0.30) and deleting any item lowers or leaves α unchanged, so every item belongs to the scale. α = 0.94 and Spearman-Brown = 0.95 indicate **excellent internal consistency**; the rank-based split-half is lower (0.85) because the half-scores are bimodal (many respondents answer 1-2 or 4-5), which inflates the Pearson r. Mean inter-item r = 0.72 is above the usual 0.15-0.50 range, so some items are probably redundant and the scale could be shortened. The other three blocks in the same file also show α ≈ 0.89, which suggests a strong common response style, so high alpha is not evidence of validity.

Dataset A (OR1-OR5) does not form a reliable scale: α = -1.65, mean inter-item r = -0.14, all corrected item-total correlations negative (-0.48 to -0.30), odd-even r = -0.65. Positively worded items of one construct should correlate positively, and a negative α is mathematically possible only when inter-item covariances are negative; together with the restricted range (almost all answers 5-7, ceiling effects) this points to a **data-quality or coding problem** in that file. I therefore do not base conclusions on it; it is reported for transparency.

Limitations. (1) Reliability ≠ validity: no evidence here of convergent/discriminant validity; (2) dataset B items are not my six items and their wording is undocumented; (3) Indonesian SME/digital-business respondents, not Viet Nam; (4) Likert data are ordinal (a polychoric/ordinal alpha would be a stricter robustness check) and the sample is a convenience sample of unknown design; (5) my own six items remain *un-piloted* - the protocol above (expert CVI, pretest, ≥ 30 responses and the same script *q7_reliability.py*) is the next step before using them.

Question 8 - Executive decision memo

To: Board of Directors | **Re:** Viet Nam operations | 12 October 2026

Decision: REDESIGN. Proceed with a phased expansion in Viet Nam, released in tranches against energy-cost and demand triggers, rather than one full commitment.

Claim 1 (IMF, Reuters 7 Oct 2026: energy prices, record public debt and AI-investment risks threaten global growth). This is a forward-looking warning in the news, not a Viet Nam finding, and our data cannot test an energy shock directly because no energy-price series was analysed. Historically, Viet Nam's GDP growth averaged 6.3% in 2000-2024 with the lowest volatility of the three countries (SD 1.35 pp vs Philippines 3.36, India 3.03), yet it fell to 2.9% in 2020 and 2.6% in 2021. In our regression, inflation and investment explained almost none of annual growth ($R^2 = 0.003$, $p = 0.96$), so we found no statistical evidence that domestic price pressure alone drives growth; this is association, not causation, from 30 points. Exports equal 98% of GDP (2025), so a global slowdown is the more plausible channel, which we did not model.

Claim 2 (IEA Global EV Outlook 2026, as reported: Viet Nam's EV share of new-car sales reached about 40% in 2025, with far higher shares projected for 2035). We cannot test sales shares with our data, and the 2035 figure is a projection, not an observation. As a loose proxy only, electricity use per capita grew 7.1% a year in 2014-2024, consistent with, but not proof of, rising power demand.

Risk indicators. (1) Inflation tail risk: Viet Nam's CPI inflation failed the Lilliefors normality test ($D = 0.211 > 0.161$) because of a 23.1% spike in 2008; it exceeded 4% in 14 of 30 years, although volatility since 2015 is low (SD 0.9 pp; 3.3% in 2025). (2) Export uncertainty: 2026 exports are forecast at US$ 502-518 bn with a 95% interval of roughly 447-557 bn.

Contingency. Stage capital in two tranches; release the second only if 2026 exports stay above US$ 447 bn and inflation below 5%; use energy-indexed supplier and customer contracts and hold 4-6 weeks of critical inputs.

Limitations. Annual data cannot capture within-year shocks; ILO labour figures are weighted survey estimates; and the news claims were assessed from published summaries, not the full Reuters and IEA texts.

(Memo body word count: 365; target 300-450.)

Appendix - AI-use disclosure and verification

Tool. An AI coding assistant (Claude Code) was used to write the Python scripts, run the analysis, build the workbook and draft this report. **Prompt (verbatim, translated from Indonesian):** 'Please do the assignment above, and I don't have the data so you find it for me' (with the exam PDF attached; follow-up: NPM 120510253003). **Data provenance:** all data were fetched by the scripts directly from the official World Bank and ILOSTAT APIs; no dataset was generated or typed by the AI. Two public Zenodo survey datasets for Q7 were located by AI-assisted web search and downloaded by script. News claims were retrieved through web search of syndicated Reuters text and IEA-based coverage because the primary pages (reuters.com, iea.org) returned access errors (401/403) to the tool.

Verification steps (to be completed and signed off by the student before submission): (1) re-run *fetch_wb.py* and *analysis.py* and confirm identical results; (2) spot-check at least three data values against the World Bank and ILOSTAT web pages; (3) confirm that the Excel formulas in sheets Q1-Q3 reproduce $D = 0.2113$ and the χ^2 values (they were recalculated in LibreOffice and match Python); (4) read the primary Reuters and IEA pages and confirm the two claims quoted in Q8; (5) rename files with name and NPM. Known limits: Q2 and Q3 p-values are inflated by weighted counts; Q7 uses public data from other respondents (Indonesia), not Viet Nam, and dataset B has no documented item wording.